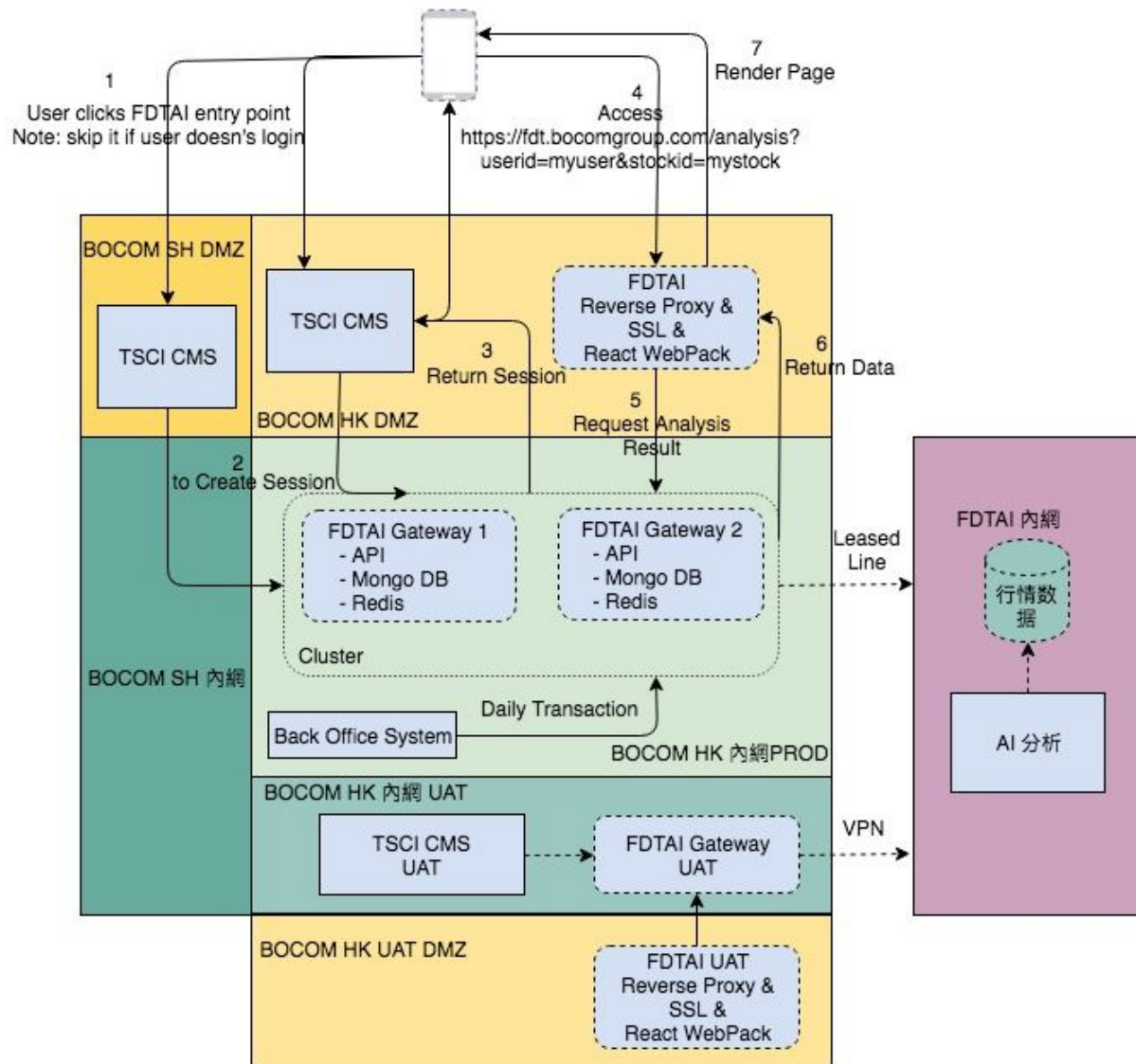
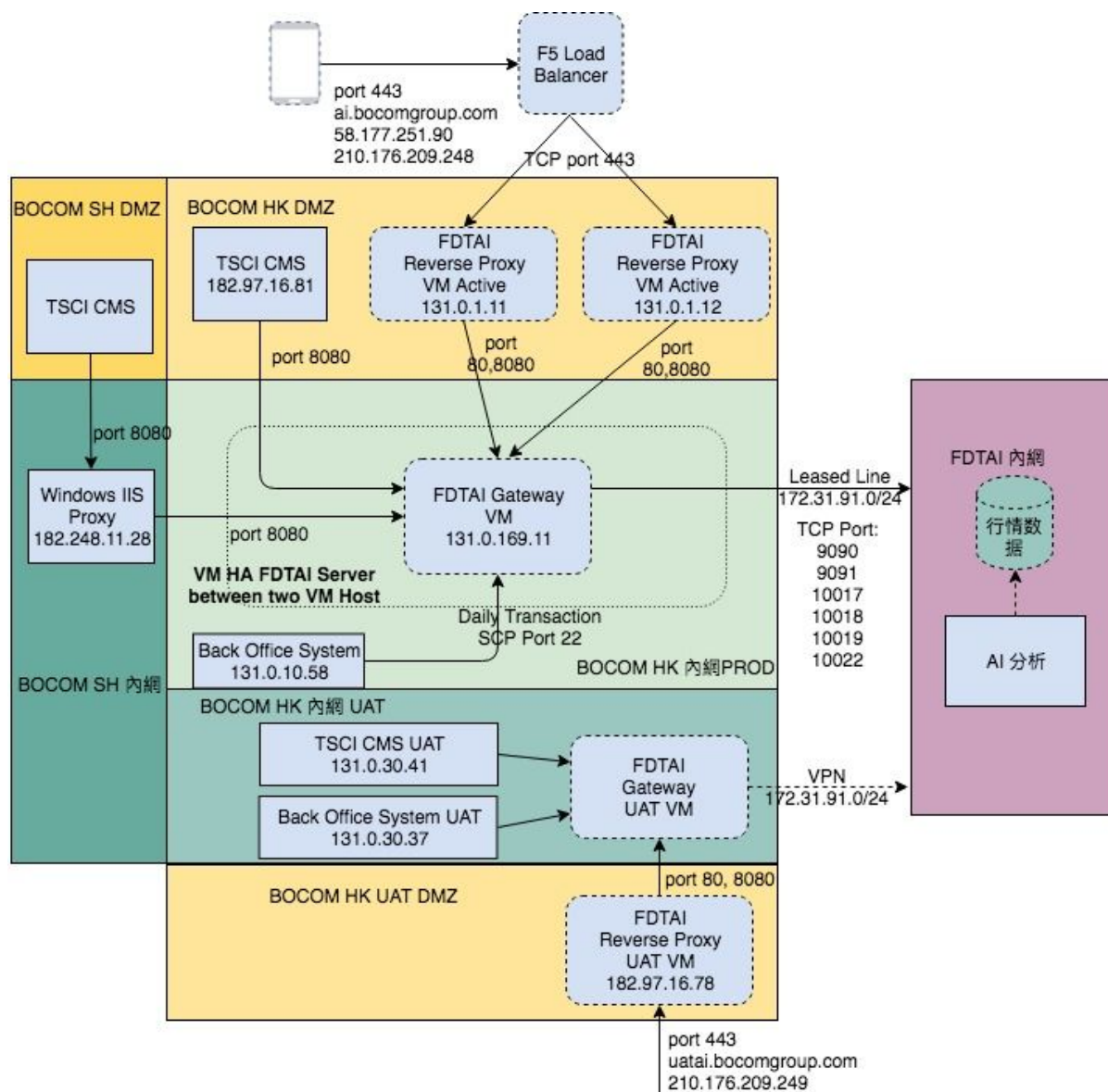


FDT-BCOM Pattern Analysis design & development

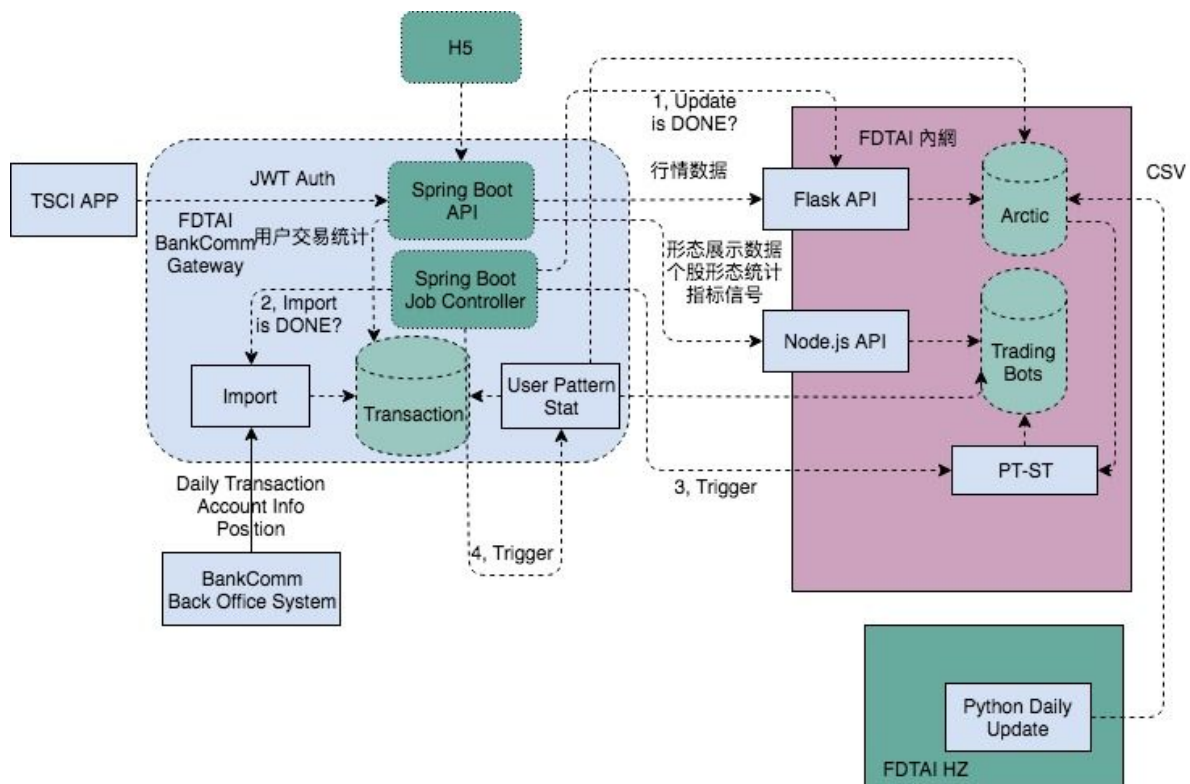
1. High level Architecture



2. Network Diagram



3. Internal API call



4. API spec between Frontend and Sprint Boot

1) Authentication

- Login - to generate JWT for logged in user to access FDTAI page.

```
curl -X POST http://47.52.174.33:8080/api/v1/login
```

```
{
  "username": "myusername",
  "goldenkey": "sun5kong"
}
{"data": "eyJhbGciOiJIUzUxMiJ9.eyJzdWwiOiJteXVzZXJuYW11IiwiaXNjaXNTE3MjgzMjU3LCJpYXQiOiJlMTcxOTY4NTd9.tZB5GLS10HdNTldhK-ARO33bcNuWX2CCY32W_nszH207sbgNQrMPIUBnxmOGRIXDnIAQhTkH76bvU99WuwF9Hw"}
```

- Logout - to invalidate JWT

```
curl -X POST http://47.52.174.33:8080/api/v1/logout -d '{"username": "183001045"}' -H
```

```
'Content-Type: application/json' -H 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWwiOiJteXVzZXJuYW11IiwiaXNjaXNTE3MjgzMjU3LCJpYXQiOiJlMTcxOTY4NTd9.tZB5GLS10HdNTldhK-ARO33bcNuWX2CCY32W_nszH207sbgNQrMPIUBnxmOGRIXDnIAQhTkH76bvU99WuwF9Hw'
{"data": "logged out"}
```

2) Stock Service - to get the market data according to the latest date in db

```
curl http://47.52.174.33:8080/api/v1/stock/instruments/000018.SZ
```

```
{
  "code": 200,
  "data": {
    "enddate": "2018-03-13",
    "ma10": [
      6.2490000000000006,
      ...
    ]
  }
}
```

```
...  
}
```

3) Pattern Analysis Service - to get the calculation result of Pattern Analysis

- Patterns list - to get the pattern list of recent 15 trading days

curl <http://47.52.174.33:8080/api/v1/patterns/instruments/00700.HK>

```
{  
  "code":200,  
  "data":[  
    {  
      "pattern_cn":"射击之星",  
      "instrument":"00700.HK",  
      "length":1,  
      "end_date":"2018-02-13T10:00:00.000Z",  
      "pattern":"ShootingStar"  
    },  
    {  
      "pattern_cn":"射击之星",  
      "instrument":"00700.HK",  
      "length":1,  
      "end_date":"2018-02-12T10:00:00.000Z",  
      "pattern":"ShootingStar"  
    }  
  ]  
}
```

- Pattern Stat - to get the pattern stat of certain stock and pattern

curl http://47.52.174.33:8080/api/v1/pattern_stat/instruments/000971.SZ/patterns/DojiStarUp

```
{  
  "code":200,  
  "data":{  
    "instrument":"000971.SZ",  
    "pattern":"DojiStarUp",  
    "best_holding_day":2,  
    "max_win_ratio":0.75,  
    "max_average_return":-0.02448203252746172,  
    "total_pattern_num":8  
  }  
}
```

- Indicators list - to get the indicator list and corresponding data for drawing

curl <http://47.52.174.33:8080/api/v1/indicators/instruments/603165.SH>

```
{  
  "code":200,  
  "data":[  
    {  
      "indicator":"RSICrossDown",  
      "time_series":[  
        {  
          "time":"2018-03-29T10:00:00.000Z",  
          "rsi12":41.73110356256725,  
          "rsi24":42.07309321925267,  
          "rsi6":42.22277613294444  
        }  
      ]  
    }  
  ]  
}
```

```

    },
    {
      "time": "2018-03-30T10:00:00.000Z",
      "rsi12": 45.457339850841414,
      "rsi24": 43.845019561506646,
      "rsi6": 49.64006467042364
    },
    {
      "time": "2018-04-02T10:00:00.000Z",
      "rsi12": 48.73564301063974,
      "rsi24": 45.44135354216762,
      "rsi6": 55.871654162544935
    },
    ...

```

- Similarities list - to get the list of stock similar to the patterns of specified stock

curl <http://47.52.174.33:8080/api/v1/similarities/instruments/600081.SH/patterns/InvertedHammer>

```

{
  "code": 200,
  "data": [
    {
      "instrument": "600519.SH",
      "instrument_cn": "贵州茅台",
      "patterns": [
        "倒锤形"
      ],
      "indicators": [
        "RSI上穿",
        "RSI下穿",
        "KDJ上穿",
        "KDJ下穿",
        "成交量均线上穿",
        "成交量均线下穿"
      ],
      "roe_cat": "高",
      "pb_cat": "高",
      "pe_cat": "中",
      "mar_cap_cat": "大盘股"
    },
    {
      "instrument": "600048.SH",
      "instrument_cn": "保利地产",
      "patterns": [
        "倒锤形",
        "阳十字星"
      ],
    },
    ...

```

- Holding list - to get the holding of specified account

curl <http://47.52.174.33:8080/api/v1/holdings/accountno/019062682>

```

"code": 200,
"data": {

```

```

"position_historical":[
  {
    "symbol":"00017.HK",
    "name":"新世界发展"
  },
  {
    "symbol":"00023.HK",
    "name":"东亚银行"
  },
  {
    "symbol":"00027.HK",
    "name":"银河娱乐"
  },
  {
    "symbol":"00175.HK",
    "name":"吉利汽车"
  },
  {
    ....

```

- User Pattern Stat - to get the user pattern stat of specified account

curl

http://47.52.174.33:8080/api/v1/user_stat/accountno/321003060/instruments/03988.HK/patterns/ShootingStar

```

{
  "code":200,
  "data":{
    "pattern_count":1,
    "user_return":"C:4.292816",
    "user_win_ratio":0.2222222238779068,
    "trade_count":9,
    "pattern_return":0.011013215859030812
  }
}

```

5. API spec between Spring Boot and Stock backend

1) to get market data

curl <http://47.52.174.33:9091/databank/bcommv1/quotes/000018.SZ>

```

{
  "code":200,
  "data":{
    "enddate":"2018-04-23",
    "ma10":[
      5.9010000000000002,
      5.9350000000000001,
      5.9170000000000002,
      5.90600000000000015,
      5.88200000000000014,
      5.8760000000000002,
      5.8760000000000002,
      5.8720000000000002,
      5.8760000000000002,
      5.9020000000000003,

```

.....

2) to get latest day k date

curl <http://47.52.174.33:9091/databank/bcommv1/dayk/lastdate>

```
{
  "code": 200,
  "data": {
    "hk_lastdate": "2018-04-20",
    "lastdate": "2018-04-20"
  }
}
```

3) to get stock's name

curl <http://47.52.174.33:9091/databank/bcommv1/names/00001.HK,00700.HK>

```
{
  "code": 200,
  "data": [
    {
      "name": "長和",
      "symbol": "00001.HK"
    },
    {
      "name": "騰訊控股",
      "symbol": "00700.HK"
    }
  ]
}
```

6. Invoking spec between Sprint Boot and Pattern Stat backend

1) indicator_signal - everyday

docker exec -it -d pattern_statistics-internal python write_to_mongo.py --action indicator_signal --date %s

2) indicator_stats - every Friday

docker exec -it -d pattern_statistics-internal python write_to_mongo.py --action indicator_stats --date %s

3) each_pattern - every Friday

docker exec -it -d pattern_statistics-internal python find_save_each_pattern.py --date %s --type update

4) technical_indicator_factory - everyday

docker exec -it -d pattern_statistics-internal python technical_indicator_to_mongo.py --date %s

7. Invoking spec between Sprint Boot and User Pattern Stat backend

1) Account Dumping - dump account csv into database

docker exec -it -d trader-pattern-stats-production python bocomaccountdump.py -i %s -t %s -c config-production.ini

2) Holding Dumping - dump holding csv into database

docker exec -it -d trader-pattern-stats-production python bocomholdingdump.py -i %s -t %s -c config-production.ini

3) Trade Dumping - dump trade csv into database

docker exec -it -d trader-pattern-stats-production python bocomtradedump.py -i %s -t %s -c config-production.ini

4) User Pattern Stat calculation - to run the calculation once all dump is done

```
docker exec -it -d trader-pattern-stats-production python bocomtraderpatternstats.py -t %s -c  
config-production.ini
```